Robustness of the SHiP muon shield to parameter tolerances

with a Taylor surrogate of the per-muon hit probability

October 24, 2026

Abstract

How much do the muon hits on the SHiP sensitive plane change when the 43 free parameters of the optimised muon shield deviate from their nominal values? Here the per-muon hit probability is modelled by a Taylor surrogate: a second-order expansion in the design parameters around the nominal design, whose gradient and Hessian are explicit coefficients, plus a residual network that is switched off at the nominal design and takes over far from it. Every conclusion is checked with the full simulation on 10^9 muons. If every parameter may be off by up to ε, the worst design found within these tolerances (searched on the surrogate, then simulated) multiplies the hits by 1.67 ± 0.07 at $\varepsilon = 1\%$, 3.6 ± 0.1 at 2%, 8.9 ± 0.3 at 3%, 103 ± 4 at 5% and 5090 ± 170 at 10%. A single parameter at its $\pm10\%$ limit gives at most $\times 2.3 \pm 0.1$ (the field of M7). Random independent deviations are much milder: harmless at $\pm2\%$, mostly harmless at $\pm5\%$, with a heavy tail at $\pm10\%$. Bounding the worst case takes sub-percent tolerances on a few parameters, mostly of M7 and M2: the allocation found for at most $\times 2$ (simulated: $\times 1.64$) holds M2's entrance width, M7's field and length and M7's inner yoke to $\pm0.5\%$. Near the nominal design every harmful direction acts through muons of 30–75 GeV, while the muons above 175 GeV respond the opposite way. The surrogate gets the directions and the worst cases right, and underestimates the magnitudes along the harmful directions by 13–20%.

Contents

1 Setup **2**

2 The surrogate **2**
2.1 Validation against the simulator 4

3 Method **5**

4 Results **6**
4.1 Single parameters 6
4.2 Curvature and active subspace 6
4.3 Worst directions 8
4.4 Concrete designs along the worst direction 9
4.5 The worst case for a given tolerance 14
4.6 Tolerance allocation 15
4.7 Random tolerances 15
4.8 Which muons respond 17

5 Conclusions **21**

A Free parameters **22**

B Integrating the surrogate over the muons **22**

C The Taylor surrogate on problems with a known answer **22**

1 Setup

The shield. The design is the optimised configuration (`try_opt`) of seven magnets M1–M7 (M0 is fixed), with 43 free parameters (Table 10): lengths `Z_len` and gaps `Z_gap`, the transverse sizes `dXIn`, `dXOut`, the inner gaps `gapIn`, `gapOut`, the yoke widths `dXYokeIn/Out` (all in cm) and the field B of each magnet (the `NI` column, the target field in tesla). Fields are uniform within each magnet block; the cavern is included.

Tolerances. Each parameter may vary by ±10% of its nominal value:

$$\phi_i = \phi_{0,i}\,(1 + 0.1\,u_i), \qquad u_i \in [-1, 1]. \tag{1}$$

$u = 0$ is the nominal design and $u_i = \pm 1$ puts parameter i alone at its limit. A step of $\|u\| = 1$ is "one tolerance unit": as large as one parameter at its limit, but spread over all of them (RMS change 1.5% per parameter). The corners of the box are at $\|u\| = \sqrt{43} = 6.6$. M6 and M7 have negative fields, so a negative change of their B lowers $|B|$.

Muons and figure of merit. Muons are drawn from the generative flow of `generator_muons` (the physical flux) and placed on axis at $z = -1$. The figure of merit H is the number of muons crossing the last sensitive plane per 10^9 muons. At the nominal design, simulator runs on the muons of the analysis give 793–852 (each ±29; the transport is stochastic) and independent muons 820±14, a hit rate of 8×10^{-7}; the surrogate gives 837. Every simulator number uses 10^9 muons, the same ones for all designs, with Poisson errors (the transport is stochastic, so this holds even on shared muons).

2 The surrogate

Model. The surrogate models the hit probability of each muon x for a design u as

$$p(\text{hit} \mid x, u) = \sigma\Big(c(x) + g(x)\cdot u + \tfrac{1}{2}\textstyle\sum_{k=1}^{8} h_k(x)\,(q_k \cdot u)^2 + \gamma(u)\,r(u)\cdot t(x) + c_0\Big), \tag{2}$$

a per-muon Taylor expansion of the logit to second order in u: the constant $c(x)$, the full gradient $g(x) \in \mathbb{R}^{43}$ and a curvature of rank 8 in a learned orthonormal basis $q_1, \ldots, q_8$ are outputs of one network of the muon (p_x, p_y, p_z, charge). The residual $r(u)\cdot t(x)$ (32 features, a DeepONet) captures what a quadratic cannot, and is switched on by the gate

$$\gamma(u) = \frac{\langle u_i^4 \rangle}{\rho^4}, \tag{3}$$

with $\langle\cdot\rangle$ the mean over the parameters and ρ the RMS radius of the training designs. The gate is zero with zero gradient and Hessian at $u = 0$, so the derivatives at the nominal design are exactly the Taylor coefficients; and it opens along a single axis (one parameter at its limit: $\gamma = 0.9$) as well as for spread-out changes, so that one-sided single-parameter responses can be represented.

Training. The Taylor coefficients are fitted to the designs near the nominal one and the residual to all: in the loss, each row's gradient reaches the Taylor part with the weight $w(u) = \exp\big(-\langle u_i^4\rangle^{1/2}/0.3^2\big)$ and the residual with weight 1 (the predictions are unchanged; only what each part learns from differs). Four such surrogates are trained with different seeds on the same data as the DeepONet surrogate and their logits averaged; the average is itself of the same form. A seed whose final validation loss is more than 10% above the median of its siblings is replaced by a new one (this happened once: 0.0189 against 0.0157–0.0162); the rule uses only the held-out training designs. Errors on surrogate quantities are the standard error of the four members.

Training data. The training data are simulated once, in eight pools (Table 1), and reused for every epoch; nothing is re-simulated during training. Each pool is a set of designs, each simulated with its own fresh muons from the flow:

- *Designs.* $u = \delta\, s\, r$, with s a scrambled Sobol point in $[-1, 1]^{43}$ and r uniform in $(0, 1)$, for a region of half-width δ ($\delta = 1$: the $\pm 10\%$ box). The radial factor r concentrates the designs towards the nominal one: of the 13 947 designs drawn this way, 4475 have every parameter within $\pm 3\%$. The first 16 designs of each base pool are the nominal design itself (8×10^8 muons at $u = 0$ in all).

- *Far from the nominal design.* The surrogate was weakest there (Table 2), so two pools were added. The *uniform* pool drops the radial factor: 1024 Sobol designs uniform in the box, all with some parameter beyond $\pm 8.5\%$ (two crashed the simulator and are missing). The *adaptive* pool (212 designs) targets where the conclusions rest and where the surrogate was least sure: the worst designs found within ± 1–10% and those of the tolerance allocations (Sec. 4.5), each with four points on its line from the nominal design and 12 perturbations within its box (24 for the one the simulator showed mispredicted, at $\pm 10\%$), and the 64 of 12 288 random designs on which the members of a first (DeepONet) ensemble, trained on the other six pools, disagreed most.

- *Muons.* Drawn from the flow at a raised *momentum* temperature T (only the momentum part of the flow is tempered), which broadens the momentum spectrum and raises the hit rate from 8×10^{-7} (the physical flux) to 1.4×10^{-4} ($T = 1.5$) and 1.8×10^{-3} ($T = 2$). This does not bias the model, which learns $p(\text{hit} \mid x, u)$ per muon; the temperature only decides where in momentum the training data is dense. The $T = 1$ pool, at the physical flux, holds only 6900 hits.

- *Rows.* Every hit and 0.4% of the misses are kept, a case-control sample; the logit is fitted with the offset $\log 250$, so the model remains one of the full probability.

- *Split.* The designs, not the rows, are split: 13 663 for training and 1518 held out for early stopping. The validation references of Sec. 2.1 are separate simulations, never used for training.

In all, 6.2×10^{10} simulated muons give 3.4×10^7 hits and 2.8×10^8 rows. At the physical flux, as many hits would need 4×10^{13} muons, i.e. about 85 000 designs of 5×10^8 muons; the same number of simulated muons would buy only ~ 120 such designs.

Table 1: Training pools. Designs are drawn in the region given, pulled towards the nominal design except in the uniform and adaptive pools (see text); designs that crashed the simulator are missing.

pool	designs	muons per design	momentum T	region	hits
main	2048	10^7	1.5	$\pm 10\%$	3.0×10^6
small region	512	10^7	1.5	$\pm 5\%$	7.0×10^5
large region	363	10^7	1.5	$\pm 20\%$	7.6×10^5
cold	512	10^7	1	$\pm 10\%$	6.9×10^3
hot	512	10^7	2	$\pm 10\%$	9.2×10^6
many designs	10 000	10^6	2	$\pm 10\%$	1.8×10^7
uniform	1022	10^7	1.5	$\pm 10\%$, uniform	1.8×10^6
adaptive	212	10^7	1.5	selected	4.0×10^5
total	15 181	6.2×10^{10}			3.4×10^7

Integration over the muons. With the muons on axis, a muon is its momentum, azimuth and charge, which are independent in the flow: (p_z, p_t) is the image of a 2-d standard normal latent. The

Table 2: Surrogate vs. simulator on the reference designs: median $|\log(\text{pred/sim})|$ (signed for the nominal design), and the relative vector errors of the 43 one-at-a-time (OAT) secants $\frac{1}{2}[H(+e_i) - H(-e_i)]$ and curvatures $H(+e_i) + H(-e_i) - 2H_0$ on the check sample. "Noise": what a perfect model would score, from the simulator's Poisson noise.

	noise	surrogate
Nominal design	1.7%	+1.3%
Designs in the half box	2.1%	8.5%
Designs in the box	1.6%	12%
Check sample: nominal design	0.2%	−1.5%
Check sample: designs in the box	0.25%	6.0%
OAT secants (cosine)	1.6%	6.4% (0.999)
OAT curvatures (cosine)	7.6%	15% (0.992)

surrogate's hits, gradient, Hessian and per-momentum splits are therefore computed on a quadrature grid (latent × azimuth × charge, App. B) instead of by sampling muons: exact to 10^{-5} in seconds, where 10^9 sampled muons leave 2–5% of noise. Only the simulator uses sampled muons.

2.1 Validation against the simulator

Reference designs were simulated on stored muons, on which the surrogate also predicts: 10^9 muons of the physical flux (the nominal design four times, 16 designs in the half box $|u_i| \leq \frac{1}{2}$, 16 in the box), and a high-momentum check sample of 4×10^8 muons drawn like the $T = 1.5$ pools (the same designs plus every parameter alone at ±10%).

On the nominal muons the surrogate predicts the nominal design to 1% and the box designs to 12%; the one-at-a-time effects on the check sample are reproduced to 6% (Table 2, Fig. 1). Within one tolerance unit the walks of Sec. 4.3 agree in direction, but along the harmful directions the surrogate is 13–20% low (see there).

The derivatives are well determined: the four members' gradients at $u = 0$ agree to cos = 0.90–0.95, and their top Hessian eigenvectors to ≥ 0.98 with the ensemble's.

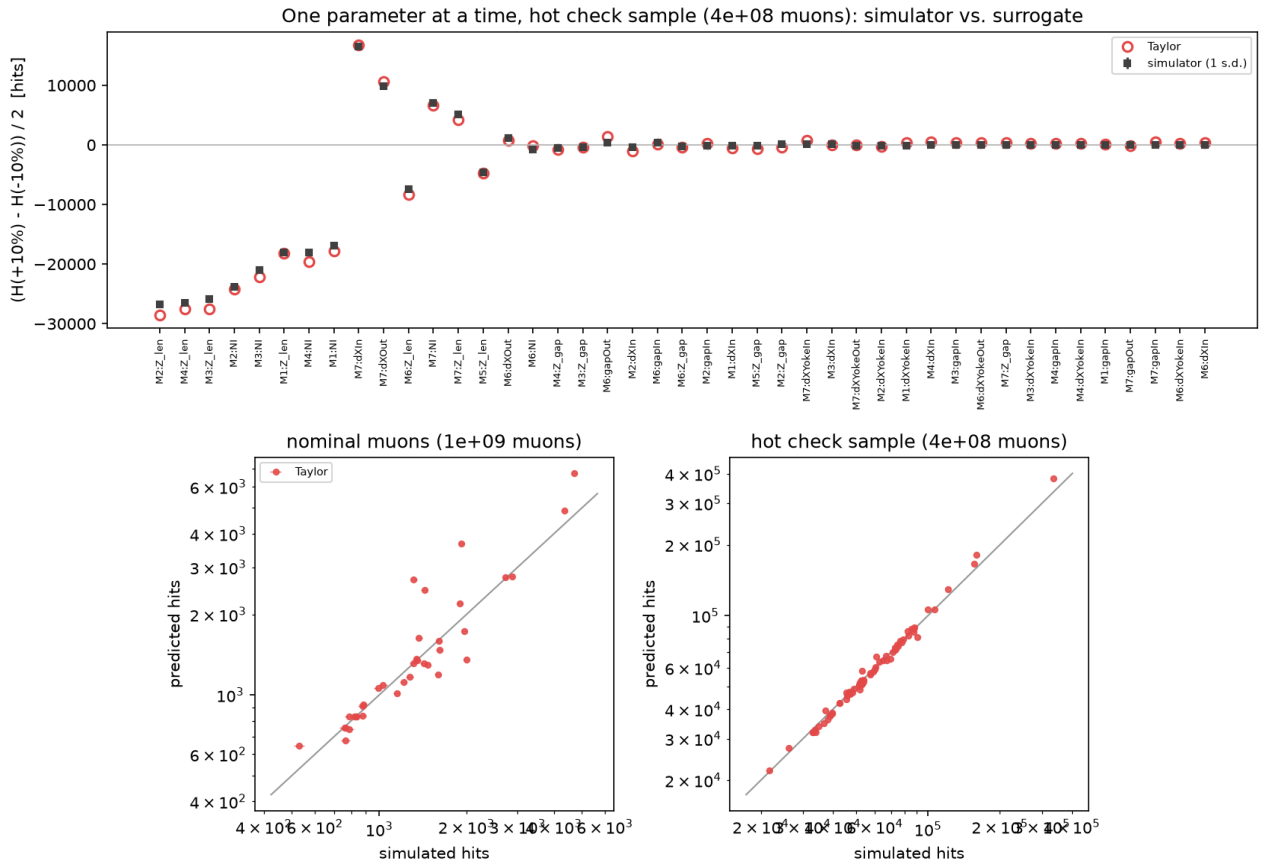


Figure 1: Top: every parameter alone at $\pm 10\%$, secant $\frac{1}{2}[H(+10\%) - H(-10\%)]$ on the check sample: simulator (squares) and surrogate (circles). Bottom: predicted vs. simulated hits of the reference designs.

3 Method

From the surrogate we compute

- the gradient $g = \nabla_u H$ and Hessian $\nabla_u^2 H$ at $u = 0$, and the exact one-at-a-time change $H(\pm e_i)/H_0 - 1$ of each parameter at its limit;
- the active subspace, the eigenvectors of $C = \mathbb{E}_x[\nabla_u p \, \nabla_u p^\top]$: the directions in which the hit probability of individual muons changes most;
- candidate worst directions, each of unit length: `ascent` ($v \propto g$), `worst` (maximiser of the quadratic model $g \cdot v + \frac{1}{2} v^\top \nabla^2 H \, v$ on $\|v\| = 1$), `hess1` (top Hessian eigenvector), `act1`, `act2` (top active directions), `corner` ($v \propto \operatorname{sign} g$: every parameter moved by the same amount) and `random`;
- the worst corner of the box, by projected-gradient ascent (`pga`);
- 4096 random designs in the box;
- the worst design for a given tolerance and a tolerance allocation (Sec. 4.5).

Each direction is then walked, $u = t\,v$ for $t \in \{-1, -\frac{1}{2}, \frac{1}{2}, 1\}$ (for `pga`, up to the box face), and simulated.

Table 3: The most sensitive parameters (surrogate): first-order sensitivity g_i/H_0 per full tolerance, and the exact change at ±10% (errors: the members' standard error).

parameter	nominal	g_i/H_0	$H/H_0 - 1$ at +10%	at −10%
M7:B	−1.9 T	−0.480	-0.30 ± 0.01	$+0.82 \pm 0.04$
M7:`dXOut`	67.81 cm	−0.340	-0.19 ± 0.01	$+0.64 \pm 0.05$
M7:`Z_len`	193.1 cm	−0.393	-0.26 ± 0.02	$+0.62 \pm 0.05$
M2:`dXIn`	68.04 cm	−0.252	$+0.05 \pm 0.01$	$+0.59 \pm 0.02$
M7:`dXYokeIn`	124.8 cm	+0.223	$+0.56 \pm 0.02$	$+0.06 \pm 0.04$
M2:B	1.9 T	−0.224	-0.13 ± 0.02	$+0.34 \pm 0.02$
M4:`Z_len`	224.0 cm	−0.138	-0.04 ± 0.03	$+0.28 \pm 0.04$
M3:`dXYokeIn`	85.88 cm	−0.094	$+0.10 \pm 0.03$	$+0.28 \pm 0.02$
M6:`Z_len`	168.0 cm	−0.173	-0.12 ± 0.02	$+0.23 \pm 0.03$
M7:`dXYokeOut`	94.92 cm	+0.079	$+0.22 \pm 0.04$	$+0.10 \pm 0.04$
M3:`dXIn`	64.81 cm	+0.111	$+0.21 \pm 0.04$	-0.03 ± 0.02
M3:`Z_len`	225.0 cm	+0.029	$+0.20 \pm 0.02$	$+0.16 \pm 0.02$
M3:B	1.9 T	−0.015	$+0.17 \pm 0.03$	$+0.19 \pm 0.03$
M2:`Z_len`	224.9 cm	+0.006	$+0.17 \pm 0.02$	$+0.18 \pm 0.01$

4 Results

4.1 Single parameters

31 of the 43 one-at-a-time effects are resolved beyond three times the members' scatter (Fig. 2, Table 3). The most sensitive parameter is the *field of M7*: lowering $|B|$ by 10% raises the hits by 82% (simulated: ×2.3). Next come the exit width and the length of M7, the entrance width of M2 and the inner yoke of M7, then the field of M2, the length of M4 and the inner yoke of M3. Most responses are one-sided. Per magnet (Fig. 3), M7 carries by far the largest share of the gradient (84% of its norm), then M2 (39%), M6 (24%) and M4 (19%).

Simulated. The eight largest were simulated at their worse limit (Table 6): all eight rise significantly. The surrogate underestimates the largest ones: the field of M7 ($\times 2.29 \pm 0.10$ simulated, 1.82 predicted), the entrance width of M2 (2.02 ± 0.09 vs. 1.59), the exit width of M7 (1.83 ± 0.08 vs. 1.64) and the inner yoke of M3 (1.49 ± 0.07 vs. 1.28); the length of M7, the inner yoke of M7, the field of M2 and the length of M4 agree within 1σ. The scans of seven of these parameters were simulated at ±5% and ±10% (Fig. 4; the length of M4 was not): 20 of 28 points agree within 2σ. The misses are the narrowing side of the steepest responses (the field of M7, the entrance width of M2, the exit width and inner yoke of M7, the inner yoke of M3), where the simulated rise is steeper than the surrogate's: 2.22, 1.93, 1.91, 1.25, 1.44 at −10% against 1.81, 1.59, 1.64, 1.06, 1.28.

4.2 Curvature and active subspace

Two Hessian eigenvalues stand out (Fig. 5): $2.39 \pm 0.04\, H_0$ per unit^2 and $1.40\, H_0$, then 0.58 and $\lesssim 0.25\, H_0$; 8 are negative, none below $-0.04\, H_0$. Along the gradient the second-order term is $0.72\, H_0$ at one tolerance unit against $0.89\, H_0$ for the first-order one: the response is strongly non-linear at that scale.

The active subspace is effectively one-dimensional (99.3% of $\mathbb{E}[\|\nabla_u p\|^2]$). Its direction `act1` shortens M1–M4 and lowers their fields together, i.e. it changes the upstream bending power (Table 5), and it is orthogonal to the gradient ($\cos = 0.02$): individual muons are deflected most along it, but the changes cancel in the total.

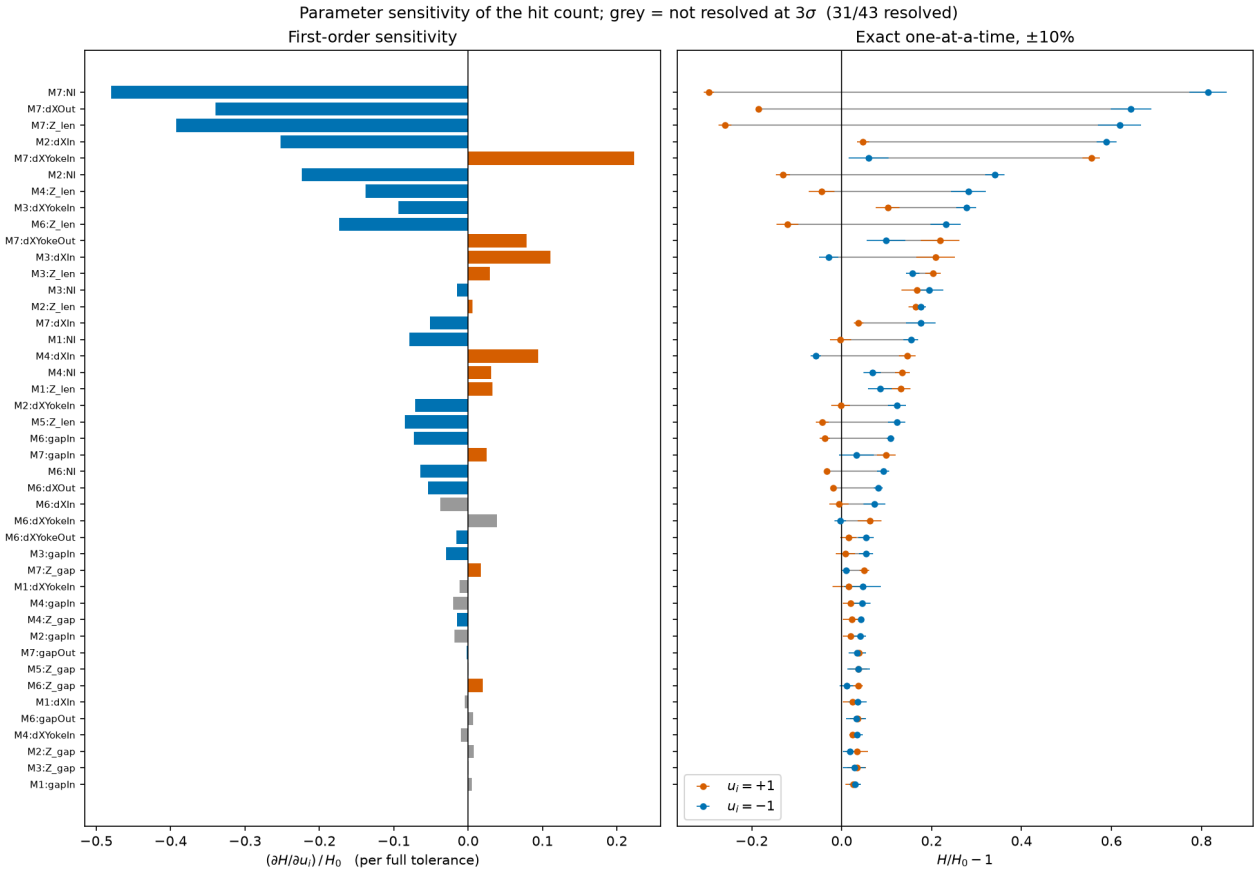


Figure 2: Sensitivity of each parameter. Left: gradient per full tolerance. Right: exact change with the parameter at +10% (orange) and −10% (blue).

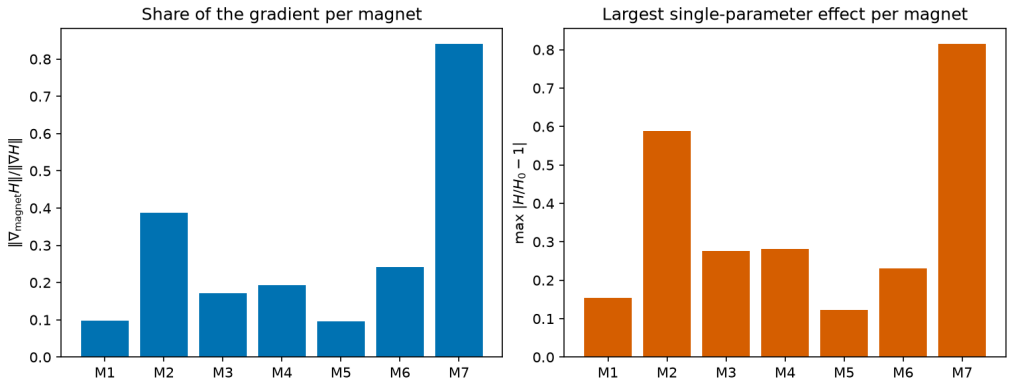


Figure 3: Per magnet: share of the gradient norm and largest single-parameter effect.

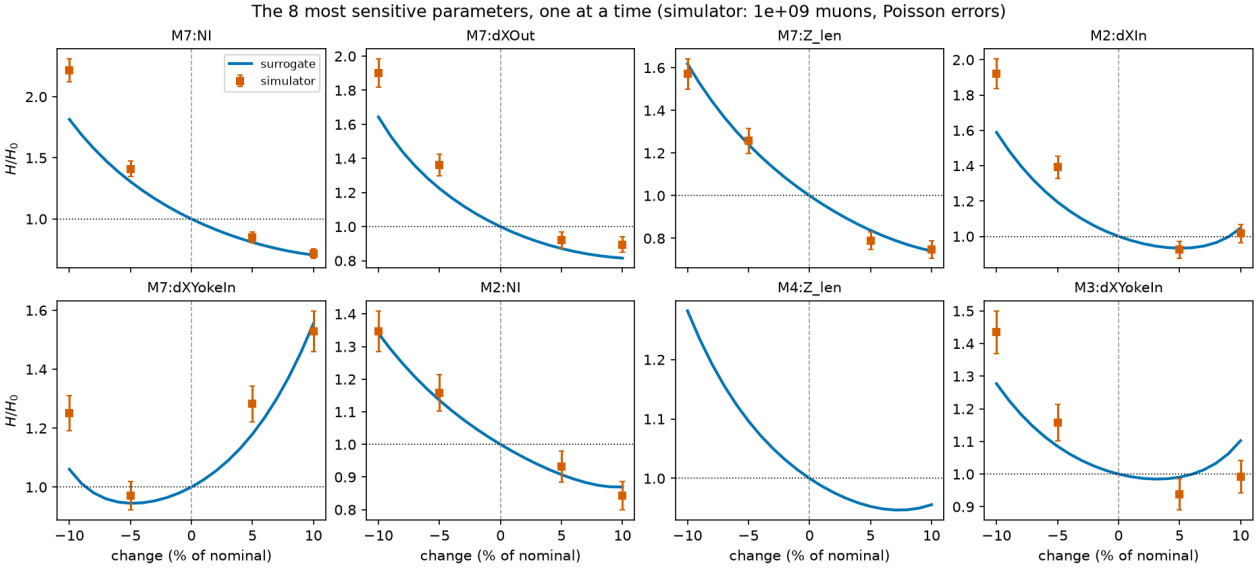


Figure 4: One-at-a-time scans of the eight most sensitive parameters: surrogate (blue) and simulator at ±5% and ±10% (orange; the length of M4 was not simulated).

4.3 Worst directions

Small coordinated changes are far more harmful than a large single one. Moving every parameter by at most 4.5% (RMS 1.5%) along `worst` multiplies the hits by 5.2 ± 0.2 (Table 4). The simplest such pattern, `corner`, moves *every* parameter by exactly 1.5% with the sign of its gradient and doubles the hits. The worst single parameter needs a 10% error for a similar effect. A random direction of the same size changes nothing (0.93 ± 0.05): the damage comes from the pattern of signs, not the size of the step.

The surrogate gets the directions; it underestimates the sizes. Its two best directions (`worst` and `hess1`) are the two the simulator ranks highest, then `ascent`. Along these harmful directions the surrogate is 13–20% low at $t = 1$ (3–4 combined standard deviations), and on the other side ($t = -1$) it is within 2σ except for `ascent` (0.56 vs. 0.68); the active direction and the random one agree (Fig. 7).

The quadratic model is not enough. The Taylor part alone, built from the gradient and Hessian, gives 2.9 instead of 5.2 along `worst` (Fig. 6): the hits grow roughly exponentially, which the residual partly captures. The derivatives give the directions; magnitudes come from the simulator.

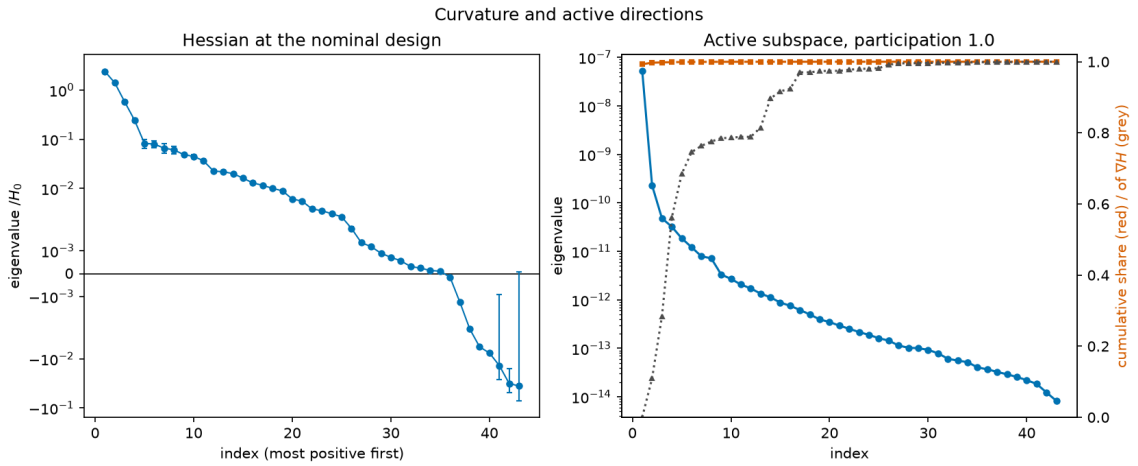


Figure 5: Left: Hessian eigenvalues in units of H_0. Right: active-subspace eigenvalues (blue), their cumulative share (red) and the fraction of the gradient they contain (grey).

Table 4: Hits one tolerance unit along each direction, surrogate and simulator (nominal: 808 simulated hits). "max / RMS": the largest and the RMS change over the 43 parameters.

direction	max / RMS change	surrogate H/H_0	simulator H/H_0	sim. hits	at $t=-1$ (sim.)
`worst`	4.5% / 1.5%	4.54 ± 0.06	5.20 ± 0.20	4204	1.05 ± 0.05
`hess1`	3.7% / 1.5%	4.35 ± 0.13	5.03 ± 0.19	4064	1.33 ± 0.06
`ascent`	5.4% / 1.5%	3.34 ± 0.08	4.09 ± 0.16	3304	0.68 ± 0.04
`act1`	4.0% / 1.5%	2.50 ± 0.06	2.47 ± 0.10	1993	1.60 ± 0.07
`corner`	1.5% / 1.5%	2.08 ± 0.05	2.24 ± 0.09	1808	0.80 ± 0.04
`act2`	8.5% / 1.5%	1.42 ± 0.01	1.67 ± 0.07	1352	0.87 ± 0.04
`random`	— / 1.5%	0.90 ± 0.03	0.93 ± 0.05	753	1.09 ± 0.05

The active subspace is a direction of curvature, not of slope. `act1` gives $\times 2.47$ at $t=+1$ and $\times 1.60$ at $t=-1$: shortening or lengthening magnets 1–4 together, with their fields, is harmful either way, and the nominal design sits at the balance point (Sec. 4.8). In the plane of `worst` and `act1` (Fig. 8) the nominal design lies in a shallow valley.

4.4 Concrete designs along the worst direction

Table 6 lists concrete designs, all simulated. "Worst, max ε" is the worst direction scaled so that no parameter moves by more than ε. It lies within the $\pm\varepsilon$ tolerances, so it bounds their worst case from below; most of its parameters stay well inside them, and the worst design within the tolerances is much worse (Sec. 4.5).

Exponential growth along the direction. Along the worst direction (Fig. 9) the hits grow by $\times 1.3$ at a maximum change of 1%, $\times 1.8$ at 2%, $\times 2.8$ at 3%, $\times 6.8$ at 5% and $\times 57$ at 10%: a factor e per $\sim 2.5\%$ of tolerance. The surrogate is 5–17% low up to 5% and nearly twice too high at 10%, beyond its reliable range. The opposite direction is harmless at 5% ($\times 1.2$) but not at 10% ($\times 4.3$), through other muons (Sec. 4.8).

The rise is collective. The largest component alone (the field of M7 lowered by 10%) gives $\times 2.2$; the top three (adding a shorter M7 and a narrower M2 entrance) $\times 6.4$; the top five $\times 13$; the top ten $\times 29$; all 43 $\times 57$ (Fig. 10). Each parameter contributes a moderate factor, and the factors multiply.

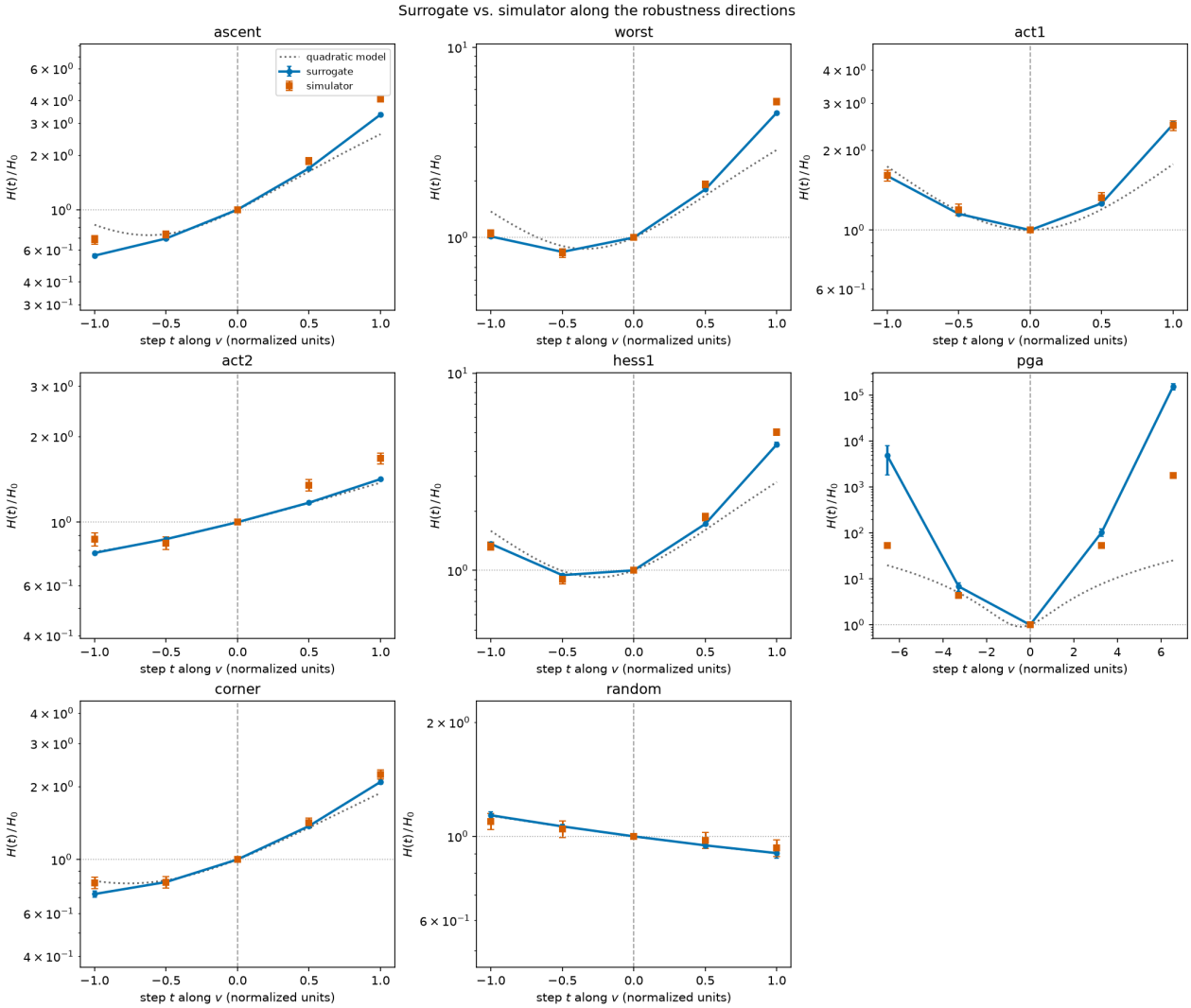


Figure 6: Walks along each direction: surrogate (blue; error bars: the members), simulator (orange) and the quadratic model (dotted).

Table 5: What each direction changes at one tolerance unit: its largest parameter changes, relative and in physical units (cm, or T for B).

direction	largest changes	in words
`worst`	M7:B −4.5% (|B| −0.09 T), M7:`Z_len` −3.7% (−7.2), M2:`dXIn` −3.7% (−2.5), M7:`dXYokeIn` +3.1% (+3.9), M7:`dXOut` −2.7% (−1.8), M3:`dXYokeIn` −2.2% (−1.9), M3:B +2.2% (+0.04 T)	weaker and shorter M7 with wider yokes, narrower M2 entrance
`hess1`	M2:`dXIn` −3.7% (−2.5), M7:B −3.6% (|B| −0.07 T), M3:B +3.1% (+0.06 T), M7:`dXYokeIn` +3.1% (+3.8), M7:`Z_len` −3.0% (−5.8), M2:`Z_len` +2.8% (+6.3), M3:`dXYokeIn` −2.5% (−2.2)	as `worst`
`ascent`	M7:B −5.4% (|B| −0.10 T), M7:`Z_len` −4.4% (−8.5), M7:`dXOut` −3.8% (−2.6), M2:`dXIn` −2.8% (−1.9), M2:B −2.5% (−0.05 T), M7:`dXYokeIn` +2.5% (+3.1), M6:`Z_len` −1.9% (−3.3)	the gradient: M7 weaker, shorter and narrower
`corner`	all 43 parameters by exactly ±1.5%, each with the sign of its gradient	the simplest uniform pattern
`act1`	M2, M3, M4:`Z_len` −3.8 to −4.0% (−8.5 to −8.9), M2:B −3.4%, M3:B −3.2%, M4:B −2.8% (−0.05 to −0.07 T), M1:`Z_len` −2.6% (−3.9)	magnets 1–4 shorter and weaker: less bending upstream
`act2`	M7:`dXOut` −8.5% (−5.7), M6:`Z_len` −2.0% (−3.3), the rest below 1.6%	the exit width of M7

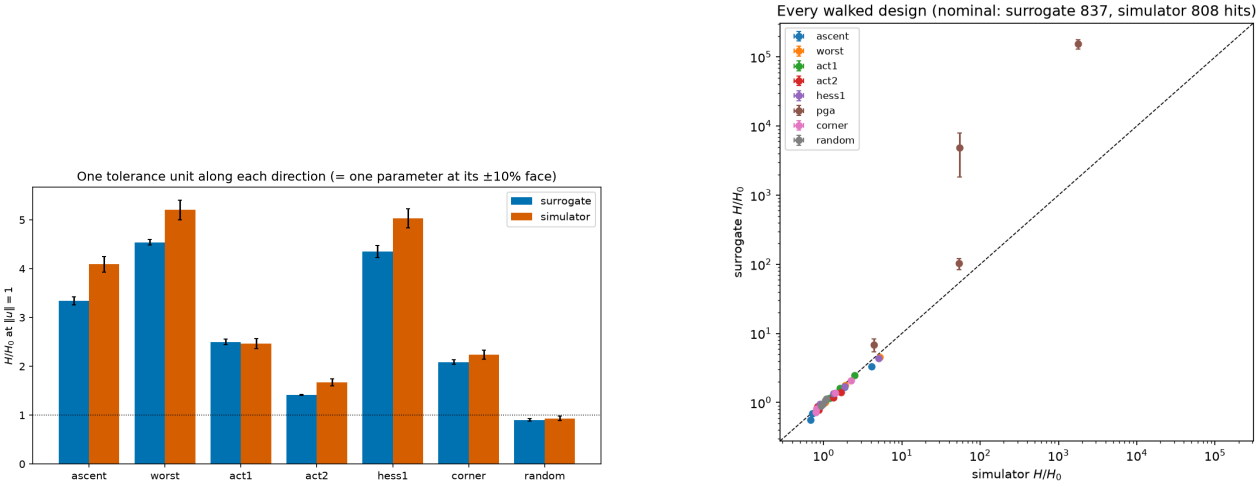


Figure 7: Left: H/H_0 at one tolerance unit along each direction. Right: surrogate vs. simulator for all 33 walked designs; the outliers are the box designs of Sec. 4.5.

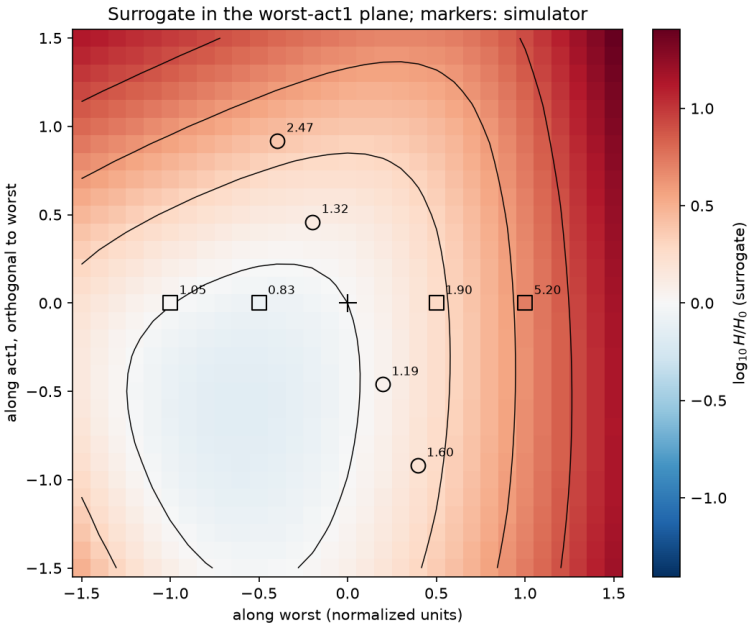


Figure 8: The surrogate in the plane of `worst` (x) and the part of `act1` orthogonal to it (y); markers: simulated designs with their H/H_0.

Table 6: Concrete designs, simulated (nominal 793 ± 28 hits) and predicted. At $\varepsilon = 10\%$ the worst direction changes $|B|$ of M7 from 1.9 to 1.71 T, M7:`Z_len` by -15.9 cm, M2:`dXIn` by -5.6 cm, M7:`dXYokeIn` by $+8.6$ cm, M7:`dXOut` by -4.0 cm, M3:`dXYokeIn` by -4.2 cm, M3:B by $+0.09$ T, M2:`Z_len` by $+9.8$ cm, M7:`dXYokeOut` by $+3.6$ cm, M4:B by $+0.07$ T, and 33 more parameters by less than 4%.

design	what changes	hits	sim. H/H_0	surr. H/H_0
worst, max 1%	all 43, RMS 0.33%	1032	1.30 ± 0.06	1.23
worst, max 2%	all 43, RMS 0.67%	1462	1.84 ± 0.08	1.64
worst, max 3%	all 43, RMS 1.0%	2211	2.79 ± 0.12	2.33
worst, max 5%	all 43, RMS 1.7%	5407	6.82 ± 0.26	5.65
worst, max 10%	all 43, RMS 3.3%	44871	56.6 ± 2.0	104
opposite, max 5%	all 43, signs reversed	917	1.16 ± 0.06	1.09
opposite, max 10%	all 43, signs reversed	3408	4.30 ± 0.17	4.73
worst, top 1 only	M7:B -10% ($\|B\|$ 1.9 $\to$ 1.71 T)	1716	2.16 ± 0.09	1.82
worst, top 3 only	+ M7:`Z_len`, M2:`dXIn` -8.2%	5055	6.38 ± 0.24	5.15
worst, top 5 only	+ M7 inner yoke $+6.9\%$, exit -5.9%	10495	13.2 ± 0.5	11.2
worst, top 10 only	the ten largest changes	23066	29.1 ± 1.1	46.3
single	M7:B -10% ($\|B\|$ 1.9 $\to$ 1.71 T)	1815	2.29 ± 0.10	1.82
single	M7:`dXOut` -10% (-6.8 cm)	1452	1.83 ± 0.08	1.64
single	M7:`Z_len` -10% (-19.3 cm)	1356	1.71 ± 0.08	1.62
single	M2:`dXIn` -10% (-6.8 cm)	1600	2.02 ± 0.09	1.59
single	M7:`dXYokeIn` $+10\%$ ($+12.5$ cm)	1284	1.62 ± 0.07	1.56
single	M2:B -10% (1.9 $\to$ 1.71 T)	1052	1.33 ± 0.06	1.34
single	M4:`Z_len` -10% (-22.4 cm)	1036	1.31 ± 0.06	1.28
single	M3:`dXYokeIn` -10% (-8.6 cm)	1181	1.49 ± 0.07	1.28

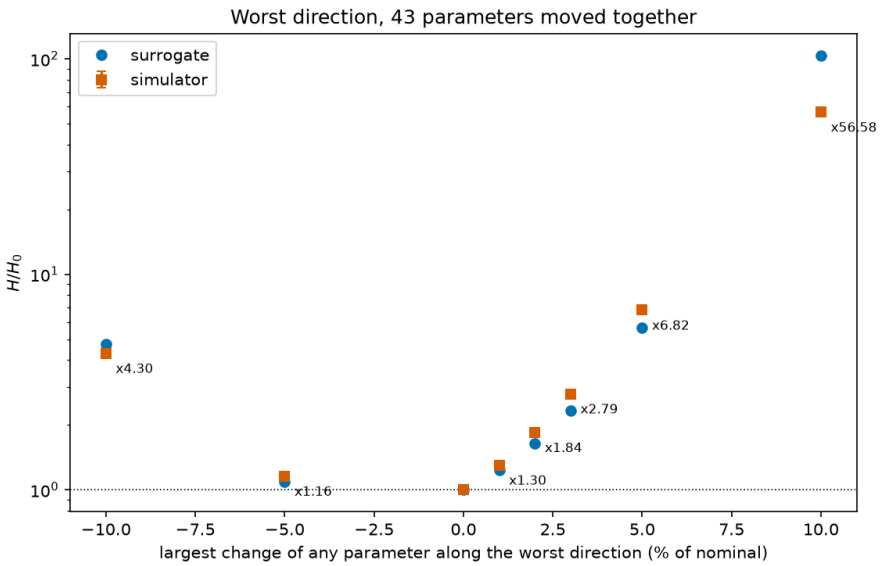


Figure 9: Hits along the worst direction against the largest change of any parameter: simulator and surrogate.

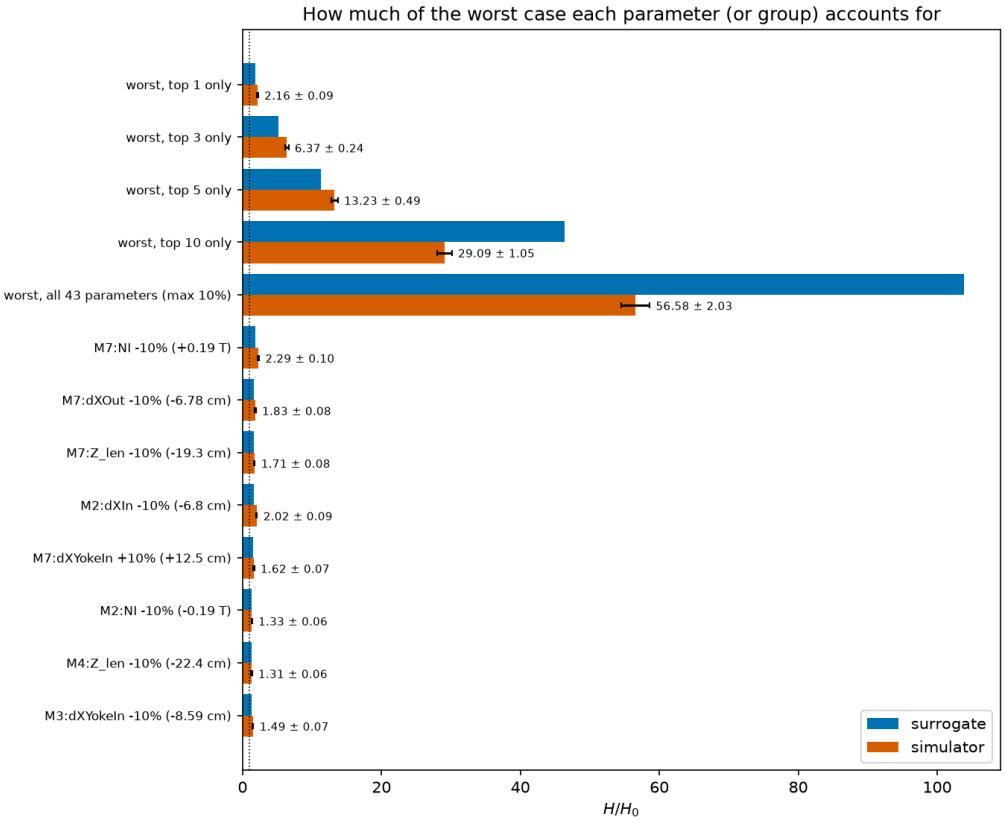


Figure 10: The worst direction at a maximum change of 10%, truncated to its largest 1, 3, 5, 10 changes, and single parameters at their worse limit.

4.5 The worst case for a given tolerance

Method. If every parameter may be off by up to ε, the designs allowed form the box $|u_i| \le \varepsilon/10\%$. Its worst design is found on the surrogate, by projected-gradient ascent of $\log H$ over the box from 16 starting points (the sign of the gradient, the scaled worst direction and random corners; 300 Adam steps each, batched), and then simulated on the 10^9 muons of the walks. The simulated values are real designs within the tolerances, so the true worst case is at least as bad; the search is as good as the surrogate. The worst direction of Sec. 4.3 maximises the hits on a sphere ($\|u\| = 1$); on a box the maximum is at a corner, every parameter at a limit with the harmful sign, so scaling the direction down underestimates it.

Table 7: The worst design within $\pm\varepsilon$ on every parameter: found on the surrogate, simulated (10^9 muons; nominal 847 ± 29). Last column: the worst direction scaled to the same ε (Table 6).

ε	at a limit	sim. hits	sim. H/H_0	surr. H/H_0	scaled direction (sim.)
1%	43	1413	1.67 ± 0.07	1.64 ± 0.02	1.30 ± 0.06
2%	43	3062	3.62 ± 0.14	3.65 ± 0.07	1.84 ± 0.08
3%	43	7550	8.9 ± 0.3	10.5 ± 0.5	2.79 ± 0.12
5%	37	86 878	103 ± 4	240 ± 60	6.82 ± 0.26
10%	37	4 311 598	5090 ± 170	$(5.2 \pm 0.3) \times 10^5$	56.6 ± 2.0

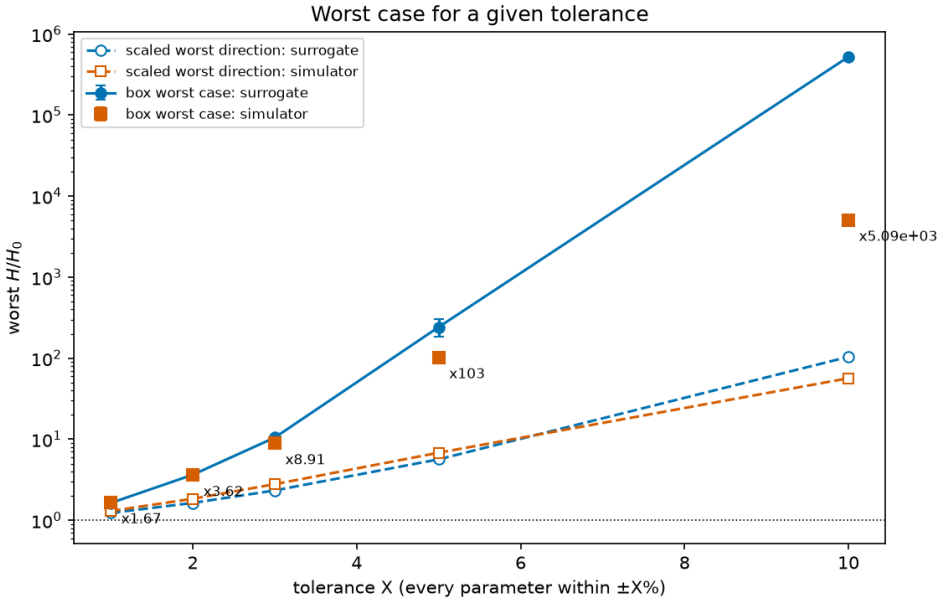


Figure 11: Worst case for a given tolerance: the worst design within $\pm\varepsilon$ (filled; surrogate and simulator) and the worst direction scaled to ε (open).

Results (Table 7, Fig. 11). Already at $\pm 1\%$ the worst design gives $\times 1.7$, at $\pm 2\%$ $\times 3.6$ and at $\pm 3\%$ $\times 8.9$, 1.3 to three times the scaled direction. Up to $\pm 3\%$ these designs act through the mechanism of the worst direction (M2's entrance narrower, M7 weaker and shorter, M7's inner yoke wider, M3's inner yoke narrower), with every parameter at its limit ($\cos = 0.65$ with the worst direction), and the surrogate predicts them to 1–18%. At $\pm 5\%$ and $\pm 10\%$ the search finds another mechanism, M7's yokes narrower with M2's entrance and M3's inner yoke wider ($\cos \approx -0.1$ with the worst direction), which the local derivatives do not show. There the surrogate overestimates by $\times 2.4$ and $\times 100$, but the designs it finds are real: $\times 103$ at $\pm 5\%$ and $\times 5090$ at $\pm 10\%$. The cruder box search of the walks (`pga`) ends at a corner with $\times 1820 \pm 60$, almost all from muons of 40–60 GeV; halfway to it the simulator gives $\times 55 \pm 2$, and its opposite corner $\times 54 \pm 2$.

4.6 Tolerance allocation

Method. Which tolerances keep the worst case below a target? Greedily: start with every parameter at ±10%; find the box's worst design u^* as above; tighten the two parameters that contribute most to it (the largest first-order shares of $\log H$, $u_i^* \, \partial \log H / \partial u_i$) by a factor 0.8, down to a floor of ±0.5%; repeat until the worst case is below the target. The first box that meets each target is its allocation, and its worst design is simulated. This is a heuristic, not an optimum.

Table 8: Tolerance allocations (greedy, surrogate) for a worst case of at most 3, 2 and 1.5 times the nominal hits: the sixteen tightest parameters (±%), and a summary of all 43. The worst design of each allocation is simulated in the last row.

parameter	nominal	$\leq 3H_0$	$\leq 2H_0$	$\leq 1.5H_0$
M2:`dXIn`	68.04 cm	0.55	0.50	0.50
M7:B	1.9 T	0.55	0.50	0.50
M7:`Z_len`	193.1 cm	0.69	0.50	0.50
M7:`dXYokeIn`	124.8 cm	0.69	0.50	0.50
M3:`dXYokeIn`	85.88 cm	0.86	0.55	0.50
M7:`dXOut`	67.81 cm	1.07	0.55	0.50
M7:`dXYokeOut`	94.92 cm	1.07	0.69	0.50
M6:`Z_len`	168.0 cm	1.34	0.86	0.50
M3:B	1.9 T	0.86	0.69	0.55
M4:B	1.9 T	1.34	0.86	0.69
M3:`dXIn`	64.81 cm	1.68	0.86	0.69
M2:`Z_len`	224.9 cm	1.34	1.07	0.69
M4:`dXIn`	39.02 cm	1.68	1.07	0.69
M2:`dXYokeIn`	70.62 cm	1.68	1.07	0.69
M2:B	1.9 T	3.28	1.34	0.69
M1:`Z_len`	151.5 cm	1.68	1.34	0.86
parameters below ±1% / ±2% / ±5%		6 / 16 / 26	11 / 19 / 33	17 / 26 / 43
median tolerance		3.3%	2.1%	1.3%
worst design: surrogate		2.97 ± 0.12	1.99 ± 0.08	1.50 ± 0.02
worst design: simulator		2.25 ± 0.09	1.64 ± 0.07	1.38 ± 0.06

Results (Table 8, Fig. 12). A guarantee against the worst case is demanding. For at most ×3, the allocation holds M2's entrance width and M7's field to ±0.55% (0.37 cm and 0.01 T), M7's length and inner yoke to ±0.7%, six parameters below ±1% and 16 below ±2%, with a median tolerance of 3.3%. For at most ×2 it puts eleven parameters below ±1%, for ×1.5 seventeen, and for ×1.25 32 (27 at the 0.5% floor, simulated $\times 1.20 \pm 0.06$); with every parameter at ±0.5% the worst case is ×1.25. The order follows the sensitivities: M2's entrance, M7 (field, length, yokes, exit width) and M3's inner yoke first, the gaps between magnets last. The simulated worst designs stay below every target (×2.25, 1.64, 1.38, 1.20): at these small tolerances the surrogate is conservative.

4.7 Random tolerances

Designs with every parameter independently uniform in ±2%, ±5% and ±10%: eight of each simulated, and 4096 per level from the surrogate (Fig. 13).

tolerance	simulated H/H_0 (8 designs)	surr. median	surr. 5–95%	$P(H > H_0)$
±2%	0.85, 0.89, 0.97, 0.98, 1.01, 1.07, 1.11, 1.20	1.03	0.87–1.23	60%
±5%	0.92, 1.01, 1.09, 1.15, 1.33, 1.45, 1.59, 4.26	1.20	0.77–2.01	74%
±10%	0.70, 1.01, 1.12, 1.35, 1.99, 3.52, 5.35, 7.99	2.03	0.79–10.1	87%

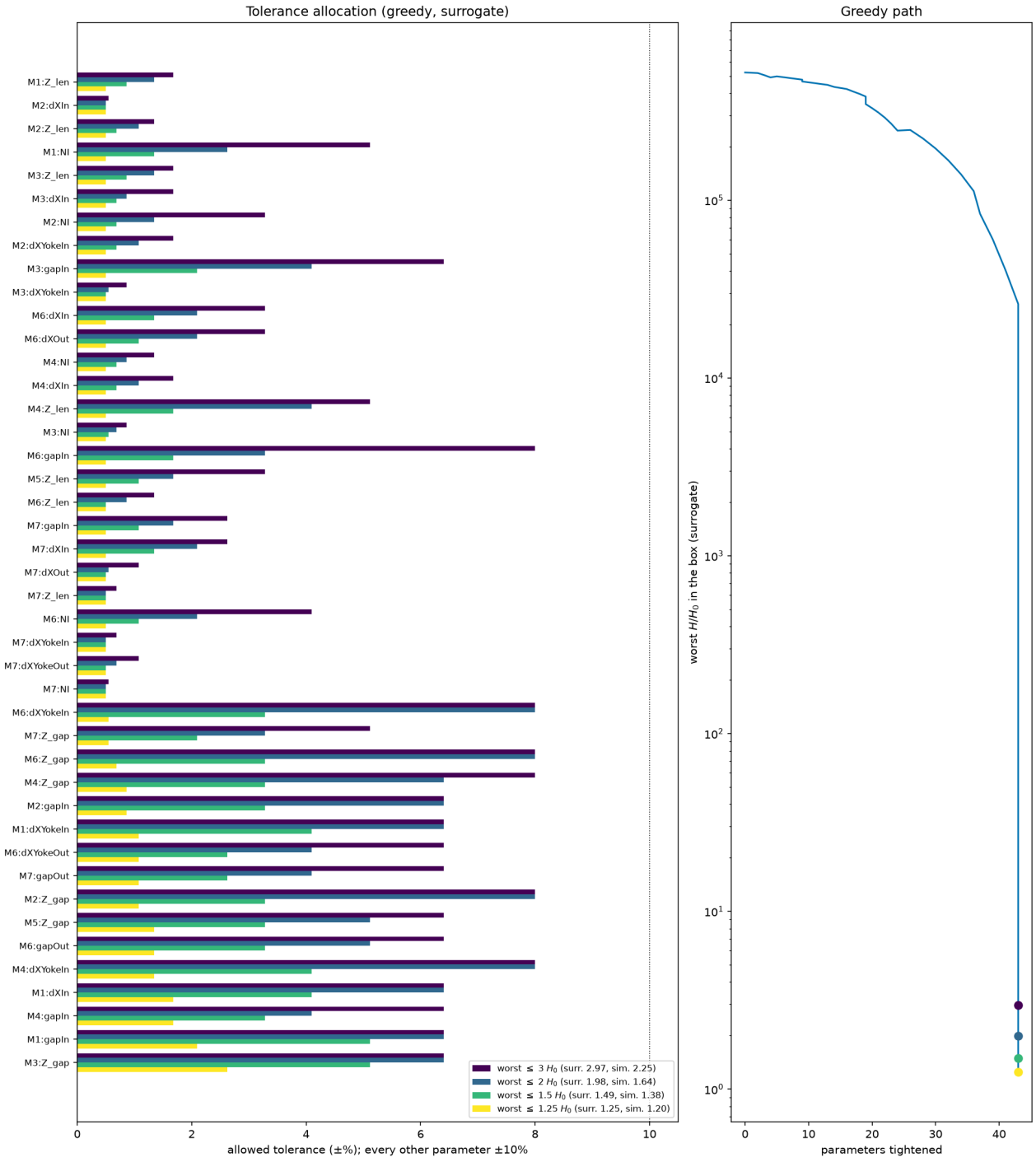


Figure 12: Left: the tolerance allocations for a worst case of at most 3, 2, 1.5 and 1.25 H_0 (surrogate; simulated worst designs in the legend). Right: the greedy path, the worst case of the box against the number of parameters tightened.

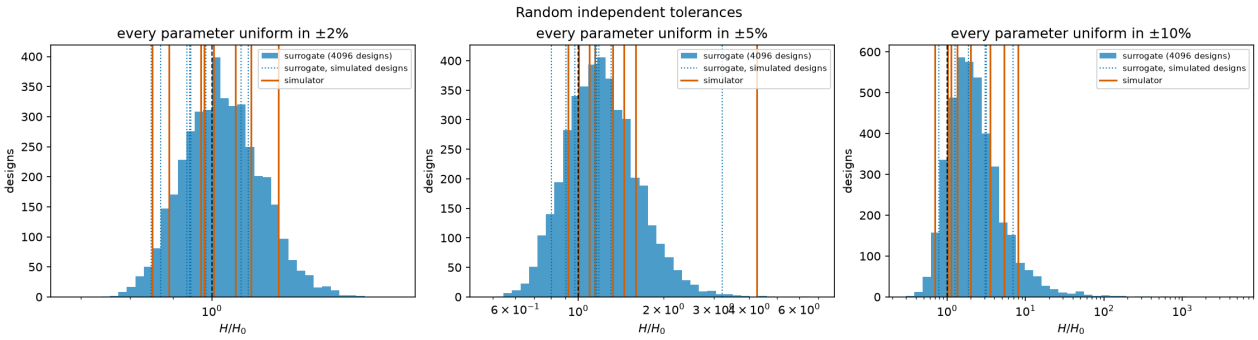


Figure 13: Random independent tolerances: surrogate distribution of 4096 designs per level (histogram), and the eight simulated designs (solid: simulator, dotted: surrogate).

Table 9: Per p_z bin at the nominal design (surrogate): share of the muons and of the hits, hit rate, relative sensitivity $|\nabla H_b|/H_b$ per tolerance unit, and the cosine between the bin's gradient and that of the total.

p_z [GeV]	muons	share of hits	hit rate	$\lvert\nabla H_b\rvert/H_b$	$\cos(g_b, g)$
10–20	1.7%	0.0%	4.3×10^{-9}	4.05	−0.37
20–30	48.4%	4.4%	7.5×10^{-8}	2.01	−0.12
30–45	28.8%	45.6%	1.3×10^{-6}	1.38	+0.90
45–75	15.3%	21.4%	1.2×10^{-6}	1.73	+0.83
75–125	4.7%	4.0%	7.3×10^{-7}	0.77	+0.64
125–175	0.89%	2.6%	2.4×10^{-6}	0.62	+0.36
175–225	0.23%	3.0%	1.1×10^{-5}	1.44	+0.05
225–300	0.08%	9.9%	1.1×10^{-4}	1.32	+0.03
> 300	0.006%	9.2%	1.2×10^{-3}	2.90	−0.01

At ±2% the hits stay within about ±20%. At ±5% most designs stay within ±60%, but one of the eight simulated quadruples the hits. At ±10% the median design is twice as bad and the tail reaches ×8 among the simulated designs (×34 at the surrogate's 99th percentile). The surrogate agrees with the simulator design by design (median ratio 1.04–1.12).

4.8 Which muons respond

The muons are split into bins of p_z (and separately p_t and charge), and each bin's hits H_b get their own gradient, one-at-a-time effects and worst direction (Fig. 15); the simulated hits per p_z range (Fig. 14) check them.

Three populations. Half of the muons have $p_z < 30$ GeV, but at the nominal design the hits come from two groups (Table 9; simulated shares in parentheses): *intermediate* momenta, 30–75 GeV, with 67% of the hits (65%; 42% at 40–50 GeV alone) at a hit rate of 10^{-6}; and *high* momenta above 125 GeV, 1.2% of the muons but 25% of the hits (24% above 150 GeV), with hit rates up to 10^{-3} above 300 GeV. A third group, *soft* muons of 20–30 GeV, gives only 4% of the hits but responds strongly to some changes.

The populations want opposite changes. The worst directions of the 30–45 and 45–75 GeV bins agree (cos = 0.92), as do all bins above 175 GeV (0.99–1.00), but the two groups are anti-aligned (cos = −0.39 to −0.52), and the soft bins are anti-aligned with the intermediate ones too (down to −0.67, Fig. 16). The total gradient, and every harmful direction of Table 4, follows the intermediate group (cos = 0.83–0.90). Simulated (Fig. 14):

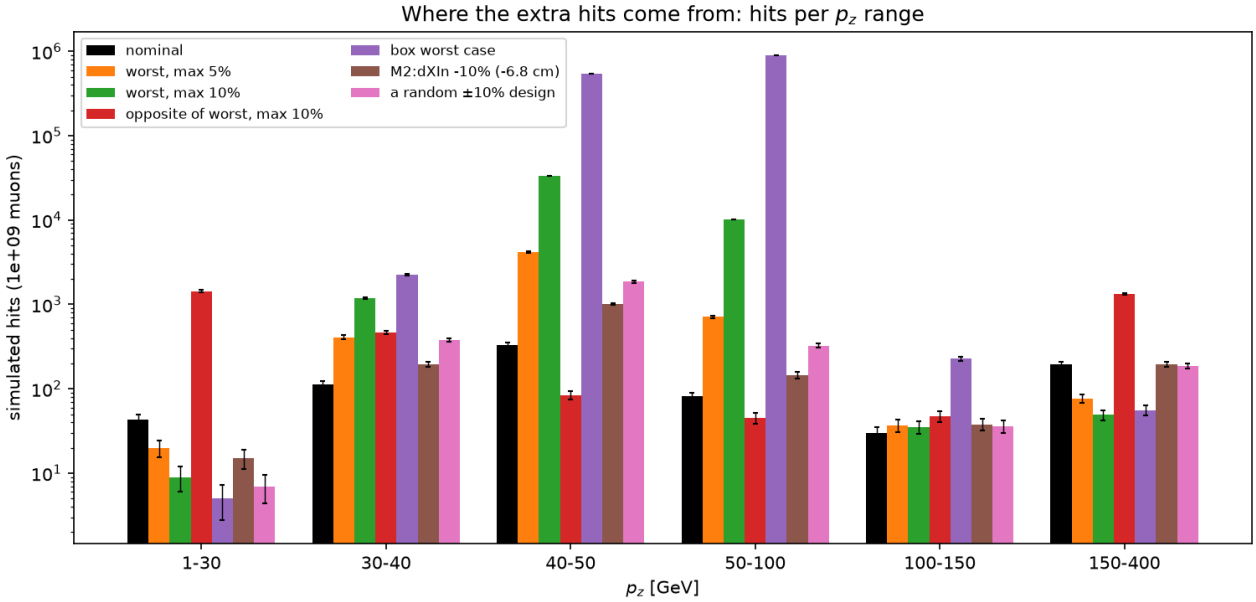


Figure 14: Simulated hits per p_z range for the nominal design and some designs of Table 6.

- worst direction at 10%: the 40–50 GeV hits go from 332 to 33 443 and the 50–60 GeV ones from 55 to 9553, while those above 150 GeV *fall* from 194 to 49 and those below 30 GeV from 43 to 9;
- opposite direction: the 40–50 GeV hits fall from 332 to 84, but those of 20–30 GeV rise from 43 to 1437 and those above 150 GeV from 194 to 1334;
- the box corner of the walks: only 30–75 GeV (516 → 1 442 724), the high-momentum hits fall (194 → 56);
- random ±10% designs can open any group: one raises the 40–50 GeV hits from 332 to 3046, another those above 150 GeV from 194 to 1066, and the worst of the eight (×8.0) both the soft (43 → 2054) and the high-momentum (194 → 3319) ones.

Which parameters act on which muons (Fig. 17). The high-momentum hits are controlled by the *upstream* magnets: the lengths of M1–M3 and the field of M3, each changing the hits above 175 GeV by 40–200% at its worse limit. The intermediate hits are controlled by the entrance width of M2, the yokes of M3 and M7, and above all the *downstream* magnet M7. The soft hits respond most to the entrance widths of M2 and M3 and the yoke of M3 (up to ×4). This explains `act1`: changing M1–M4 together shifts hits between the high-momentum group and the others, and the nominal design balances them.

Transverse momentum and charge. All p_t bins below 1.5 GeV behave alike (worst directions at cos = 0.92–1.00). The muons above 1.5 GeV (1.6% of the muons, 22% of the hits) are less sensitive (0.78 vs. 1.0 per tolerance unit) and only partly aligned (cos = 0.45–0.58, Fig. 18). Positive muons give 64% of the hits; both charges respond to the same changes (cos = 0.96), the μ^- somewhat more strongly (1.00 vs. 0.84).

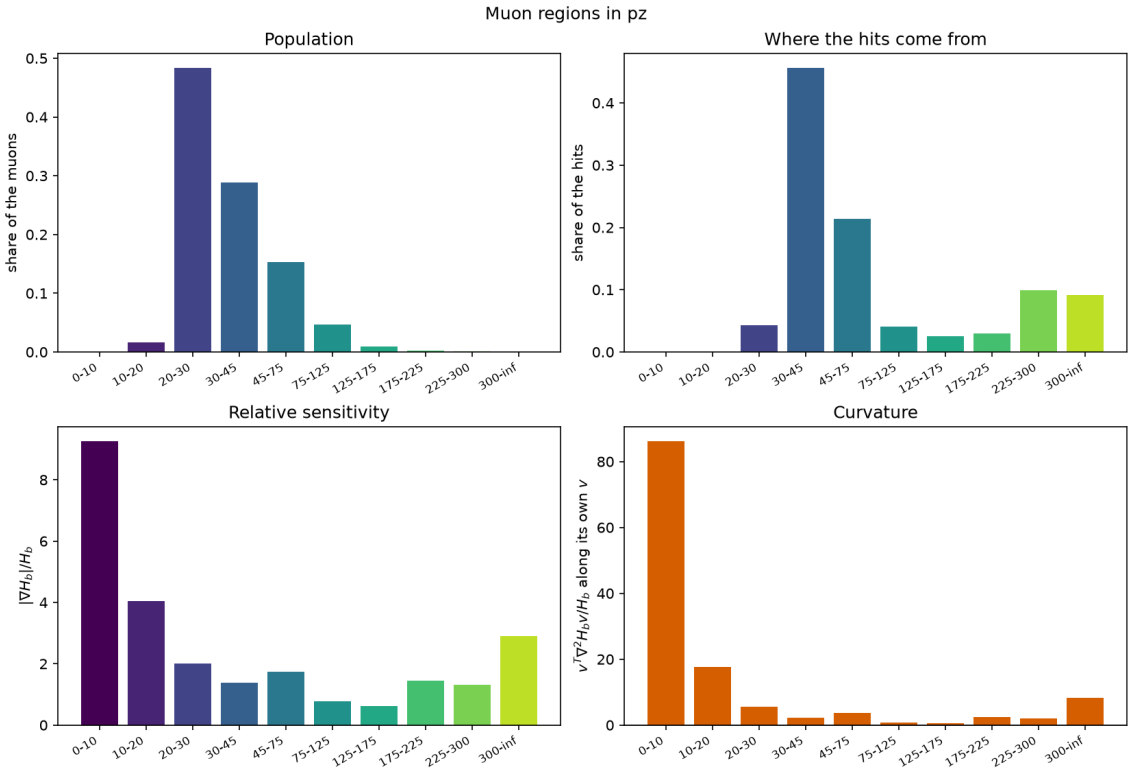


Figure 15: Per p_z bin (surrogate): share of the muons and of the hits, relative sensitivity and curvature along the bin's own worst direction.

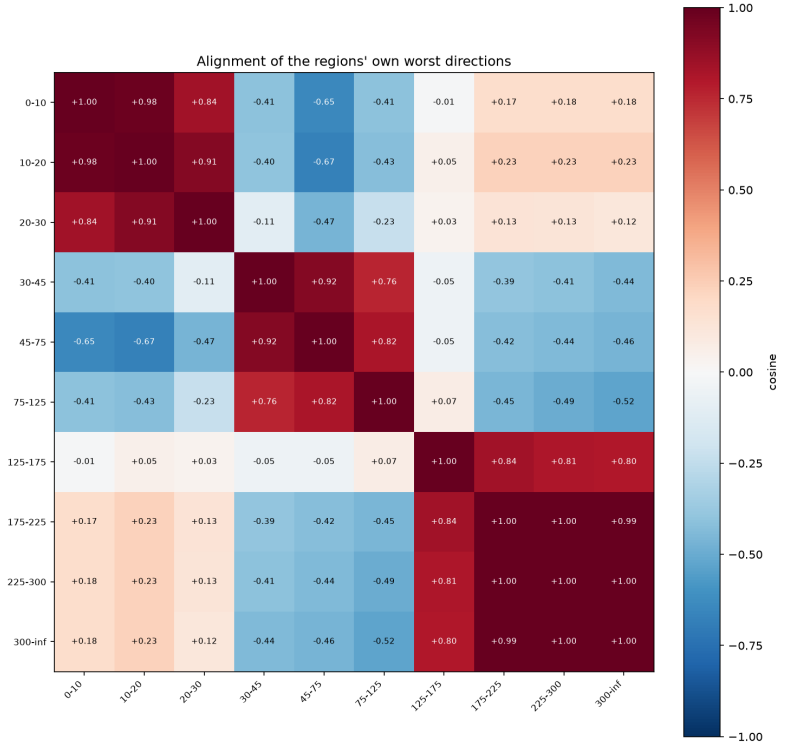


Figure 16: Cosine between the worst directions of the p_z bins.

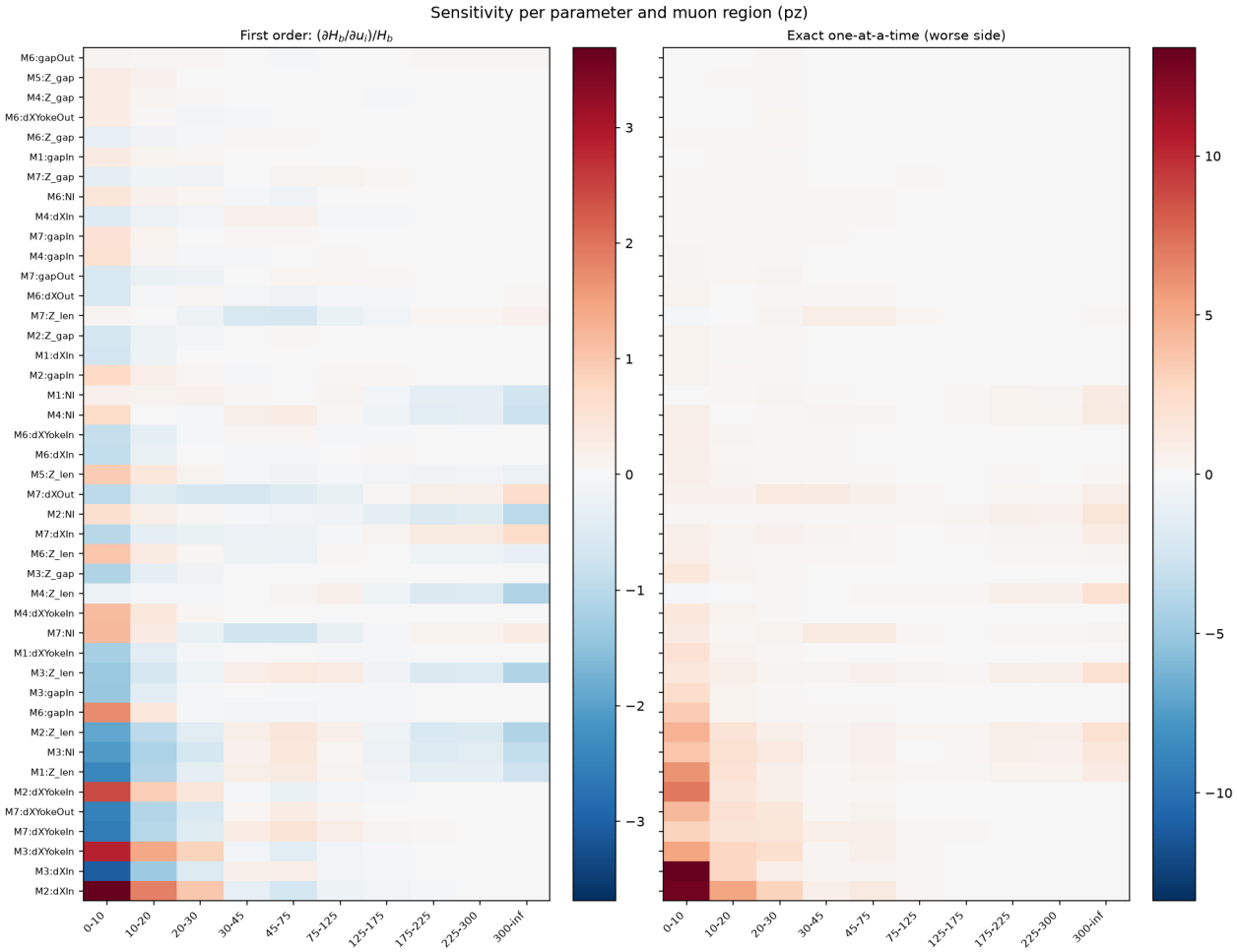


Figure 17: Sensitivity of each p_z bin to each parameter, relative to the bin's own hits. Left: first order. Right: exact one-at-a-time change at the worse of $\pm 10\%$.

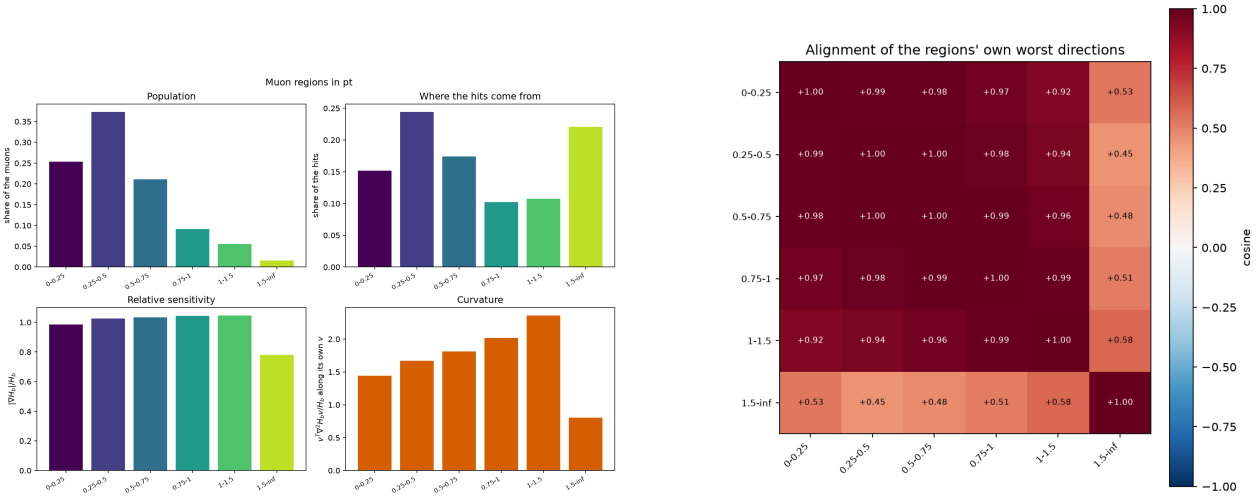


Figure 18: Per p_t bin: overview (left) and alignment of the bins' worst directions (right).

5 Conclusions

1. **The design is fragile to coordinated deviations.** With every parameter off by up to ε, the worst design found (on the surrogate, then simulated) gives $\times 1.7$ at $\varepsilon = 1\%$, $\times 3.6$ at 2%, $\times 8.9$ at 3%, $\times 103$ at 5% and $\times 5090$ at 10%; these are lower bounds on the true worst case. Moving every parameter by 1.5% with the sign of its gradient already doubles the hits.

2. **Bounding the worst case needs sub-percent tolerances on a few parameters and a few percent on most.** The allocation found for at most $\times 3$ holds M2's entrance width and M7's field to $\pm 0.55\%$ and M7's length and inner yoke to $\pm 0.7\%$, 16 parameters below $\pm 2\%$, median 3.3%; for at most $\times 2$, eleven below $\pm 1\%$ (simulated $\times 2.25$ and $\times 1.64$). M7, M2 and M3 need the tightest tolerances, the gaps between magnets the loosest.

3. **Single parameters underestimate the risk.** The worst one at its $\pm 10\%$ limit (the field of M7) gives $\times 2.3$; the entrance width of M2 and the exit width and length of M7 give $\times 1.7$–2.0. The top three changes of the worst direction together give $\times 6.4$, the top ten $\times 29$: the factors multiply.

4. **Random deviations are much milder than the worst case.** Independent errors of $\pm 2\%$ keep the hits within about $\pm 20\%$; at $\pm 5\%$ most designs stay within $\pm 60\%$ but some quadruple the hits; at $\pm 10\%$ the median design is twice as bad and the tail reaches $\times 8$–34.

5. **Different muons pull in opposite directions.** The hits come from muons of 30–75 GeV (65%) and above 150 GeV (24%). Every harmful direction near the nominal design acts through the first group, while the high-momentum hits fall; the reverse direction raises the soft and high-momentum hits instead. The dominant active-subspace direction (the lengths and fields of M1–M4) trades hits between the groups and is harmful on both sides.

6. **The Taylor surrogate gets the directions and the worst cases, not all the sizes.** Its worst directions are confirmed by the simulator ($\times 5.2$ at one tolerance unit), and its worst designs within ± 1–3% are predicted to 1–18%, but along the harmful directions it is 13–20% low and it underestimates the steepest single-parameter responses; far from the nominal design it overestimates. Every number used for a decision here is simulated.

Limitations

- Uniform fields per magnet block, no field maps.
- Muons come from the generative flow, so the absolute numbers depend on how well the flow reproduces the muon spectrum.
- The surrogate's errors are the scatter of its members: they miss a bias the members share, such as the underestimate along the harmful directions (Sec. 4.3).
- The random-tolerance distributions rest on eight simulated designs per level and on the surrogate for their tails.
- The worst designs within a tolerance are found on the surrogate: their simulated values bound the true worst case from below, possibly loosely at $\pm 5\%$ and $\pm 10\%$, where the surrogate is unreliable. The tolerance allocation is a greedy heuristic, not an optimum.

A Free parameters

Table 10: Free parameters and nominal values (lengths in cm, B in T).

magnet	Z_gap	Z_len	dXIn	dXOut	gapIn	gapOut	dXYokeIn	dXYokeOut	B
M1	–	151.5	63.12	–	8.149	–	64.12	–	1.9
M2	15.0	224.9	68.04	–	8.431	–	70.62	–	1.9
M3	16.12	225.0	64.81	–	8.986	–	85.88	–	1.9
M4	15.0	224.0	39.02	–	8.327	–	97.53	–	1.9
M5	15.0	120.4	–	–	–	–	–	–	–
M6	15.0	168.0	20.0	30.0	24.92	9.388	97.18	102.7	−1.2
M7	15.0	193.1	37.95	67.81	8.000	8.000	124.8	94.92	−1.9

B Integrating the surrogate over the muons

With $A(z)$ the kinematic envelope of the sampled muons and $N = 10^9$,

$$H(u) = \frac{N}{\int \mathrm{d}^2 z\, \mathcal{N}(z) A(z)} \sum_{q=\pm 1} P(q) \int \mathrm{d}^2 z\, \mathcal{N}(z)\, A(z) \int_0^{2\pi} \frac{\mathrm{d}\varphi}{2\pi}\, p\big(\text{hit} \mid u, x(z, \varphi, q)\big), \tag{4}$$

where z is the flow's 2-d momentum latent, φ the azimuth (uniform) and q the charge (with the data's μ^+ fraction). The derivatives use 801^2 latent points on $[-7, 7]^2$ with Gaussian weights, 64 azimuths and both charges; everything else $201^2 \times 64 \times 2$ points, with the trunk features cached. For the DeepONet surrogate the grid is converged to 10^{-5} from about 10^7 points (3×10^{-4} at 10^6), while 10^9 sampled muons scatter by up to 2% in H_0 and 5% above 300 GeV, and quasi-Monte Carlo helps only by a factor two. Importance-sampling the simulator at a raised temperature does not help either: the hits that matter are at intermediate momenta, of which a hotter sample holds fewer.

C The Taylor surrogate on problems with a known answer

On four toy problems with a binary outcome per sample and known probabilities (10–43 parameters), the exact gradient and Hessian at $u = 0$ are available. Trained on increasing numbers of simulated samples, the Taylor surrogate's errors on both fall steadily up to 10^8 samples, without the plateau of a general-purpose network; with the weighted loss used here it needs 2–12 times more samples than an unweighted fit for a given Hessian accuracy, and stays somewhat less accurate. The details, and the comparison with the DeepONet surrogate, are in the companion comparison report.